AI and Robotics

Lesson 2a: Components of a Robotic Agent

Sarah Keren

The Taub Faculty of Computer Science
Technion - Israel Institute of Technology



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Components of a Robotic Agent

Reminder: Autonomy

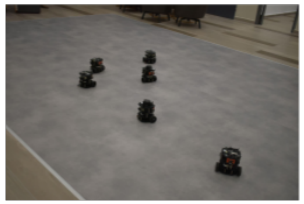
- Action (Actuation)
  - Locomotion: wheels (e.g., differential drives)
  - Manipulation: arms, grippers, etc.
- Perception (Sensors)
  - Proprioception (internal: IMU (inertial measurement unit), encoders , etc,)
  - Exteroception (external: cameras, scanners)
- Cognition (Control)
  - From reactive to proactive
  - From finite state machines to cognitive robotics

Many Types of Robots

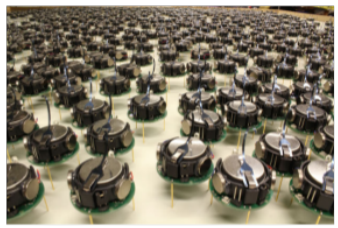
















Components of a Robotic Agents

Sensors



Controllers



Actuators

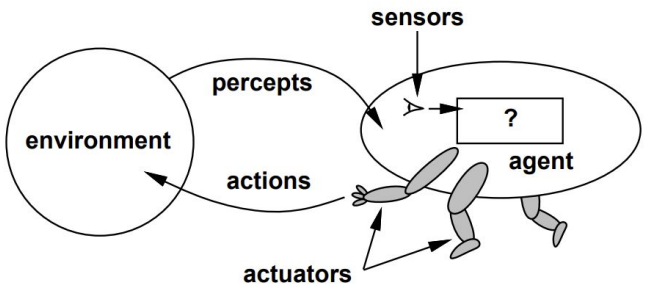




Components of a Robotic Agent

Robot State

Components of a Robotic Agents



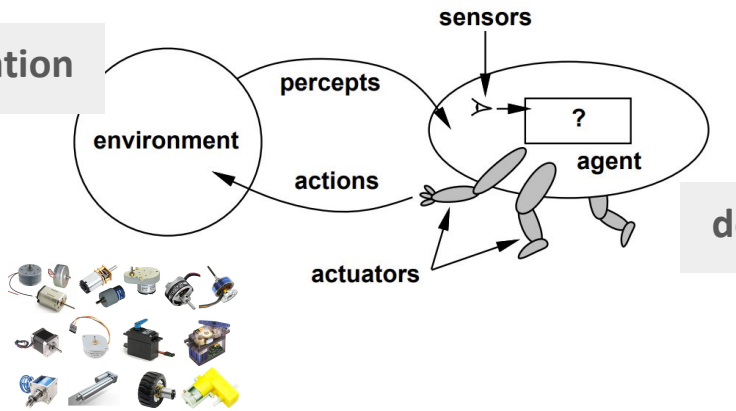
Components of a Robotic Agent

Robot State

Components of a Robotic Agents



state estimation





decision making

motion control

Actuators

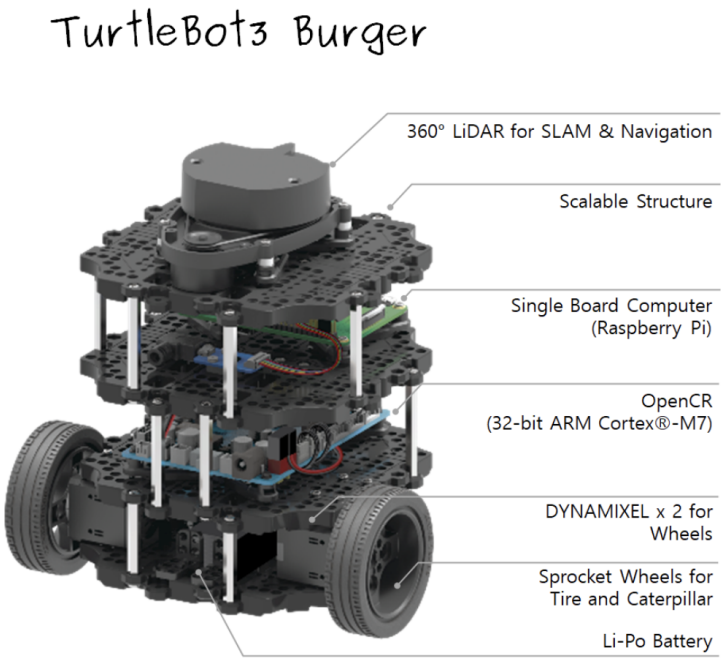
- Actuators control movements within machines.
- There are various kinds of actuators that
  - produce varying motions (linear or rotary)
  - use different power sources
    - hydraulic (with a fluid-filled cylinder)
    - pneumatic (using pressurised gases)
    - electric (popular in robotics)



https://youtu.be/LHn7O6PUaoY?t=68 www.creativemotioncontrol.com/types-of-actuators/

The turtlebot3 burger actuators

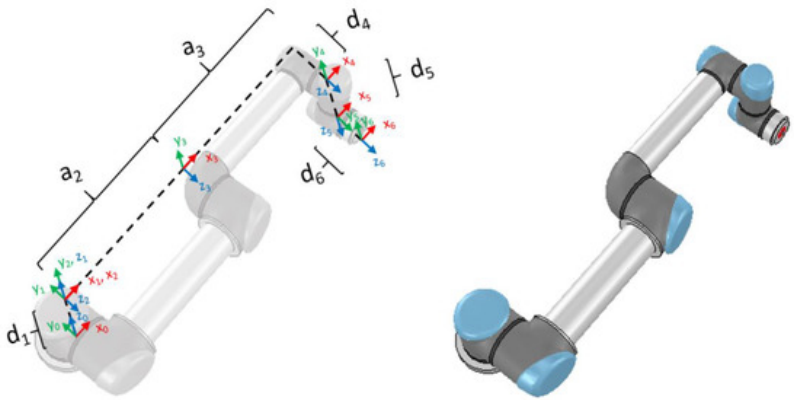
dynamixel





https://www.youtube.com/watch?v=rmqyOCefhGA https://emanual.robotis.com/docs/en/dxl/x/xl430-w250/

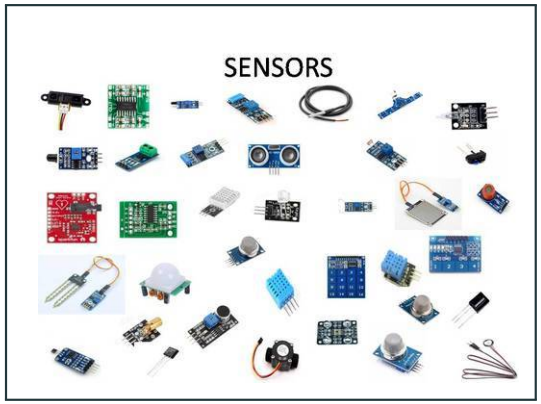
The UR actuators



- The UR robot (by Universal Robots) is a 6-degree-of-freedom (6-DoF) collaborative robotic arm.
- Its actuators are the electric servo motors integrated into each of its six joints.

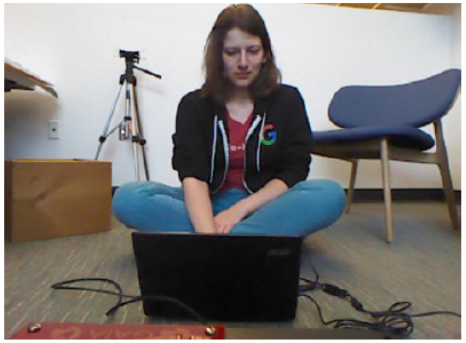
Sensors

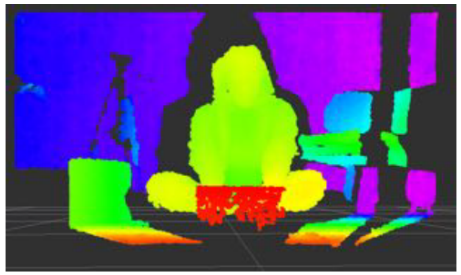
- Used to estimate the robot's condition and environment and passed to the controller to decide how to act.
- Generally classified into two groups:
  - **Internal sensors**
    - obtain the information about the robot itself (e.g., position, velocity, acceleration, motor torque sensor, etc)
  - **External sensors**
    - gather information from the surrounding environment (e.g., cameras, laser range finder, contact and proximity sensors, and force sensors)



Sensors: Vision

Many vision options: color, depth, video, and more.







Vision

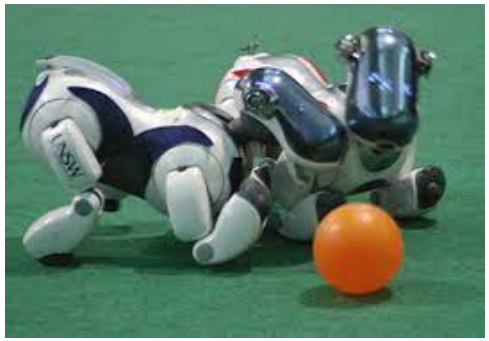
Why is vision challenging?



Vision

Why is vision challenging ?



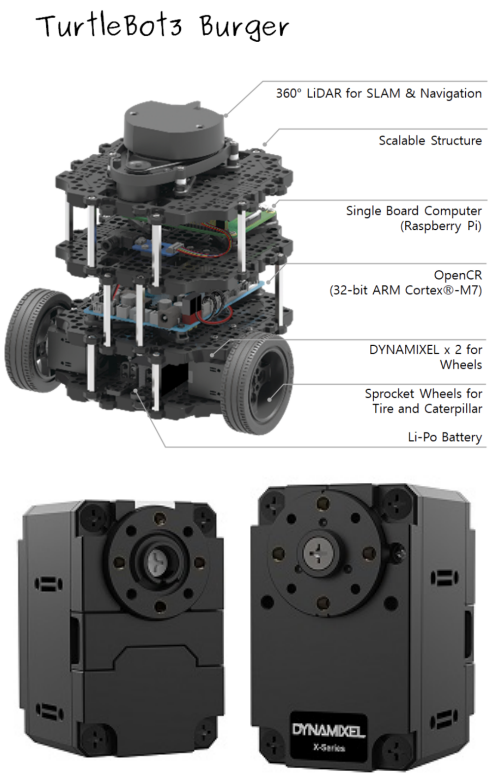




https://youtu.be/pfdNm3wFASE

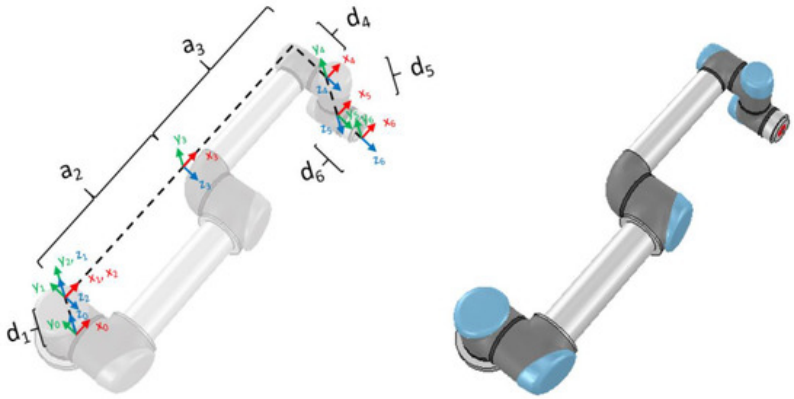
https://www.youtube.com/watch?v=so9axknlftk

The turtlebot3 burger sensors



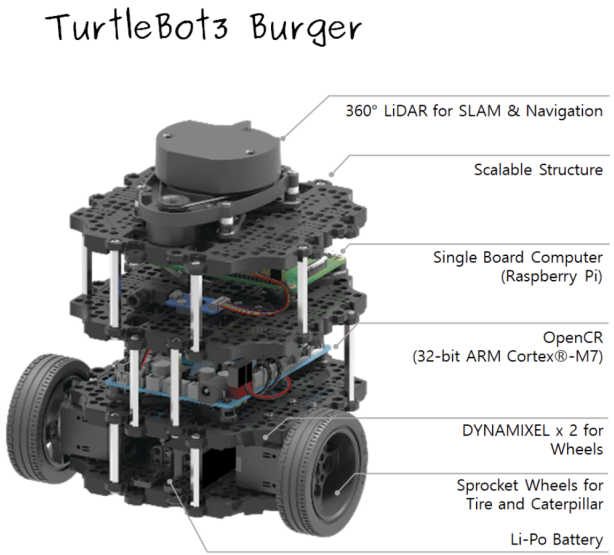
- 360 degree Laser Distance Sensor - Laser Distance Sensor LDS-01
- Gyroscope - measures or maintains orientation and angular velocity.
- Accelerometer - measures acceleration.
- Magnetometer 3 Axis - measures magnetic induction (magnetic field intensity),

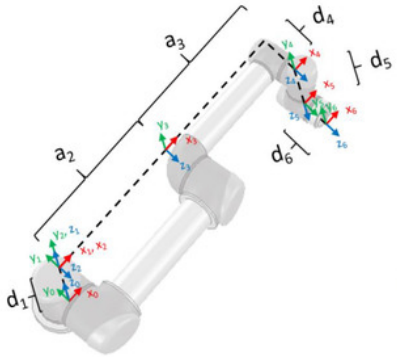
The UR Sensors

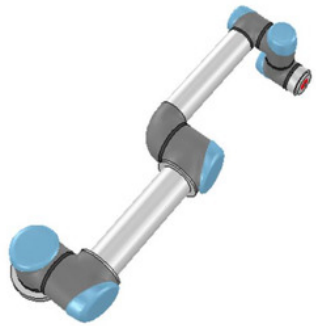


- Each actuator includes:
  - Absolute encoders - for joint angle measurement
  - Torque sensors - for collision detection and compliant motion
  - Temperature sensors- for thermal protection

Control and Decision-Making





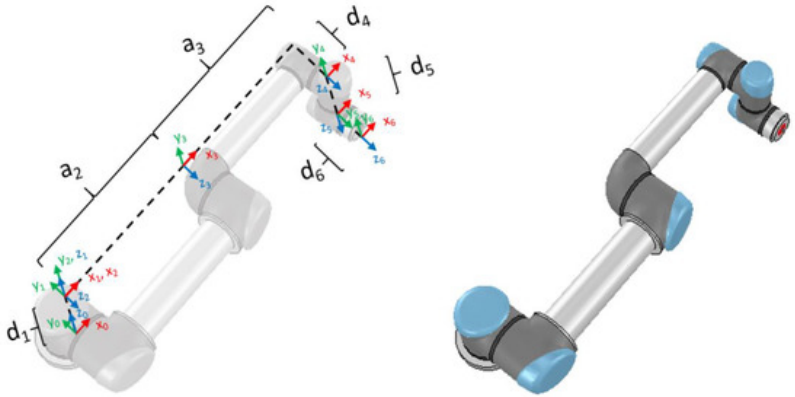


- Varies according to setting and objectives
- Highly influenced by uncertainty and the way it is accounted for
- A major focus of this course

Robot State

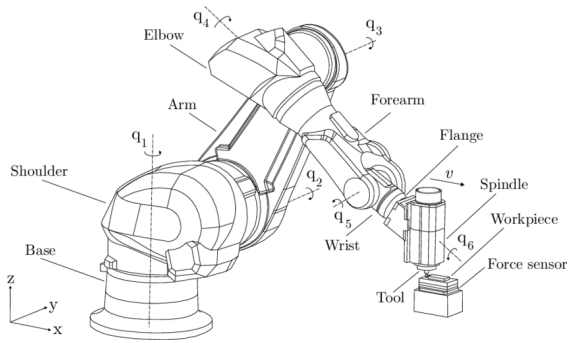
Configuration

- The **configuration** of a robot is a complete specification of the position of every point of the robot.
- A configuration q is **collision-free**, or free, if the robot placed at q does not intersect any obstacles.



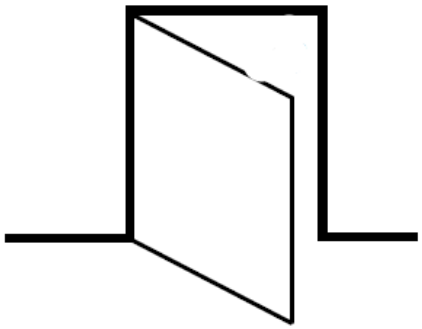
Degrees of Freedom (dof)

- The **degrees of freedom (dof)** of a robot are the minimum number n of real-valued coordinates needed to represent the configuration of a robot.
- A mechanism is typically constructed by connecting rigid bodies, called **links**, together by means of **joints**, so that relative motion between adjacent links becomes possible.
- **Actuation** of the joints, typically by electric motors, then causes the robot to move and exert forces in desired ways.
- dof typically refers to the number of movable joints of a robot.

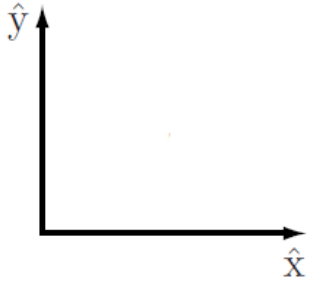


Degrees of Freedom (dof)

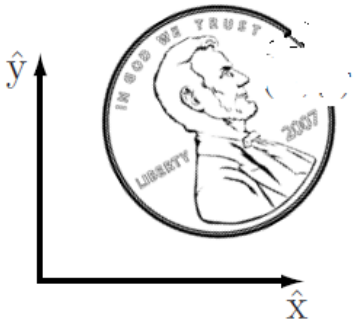
What's the dof ?



(a)

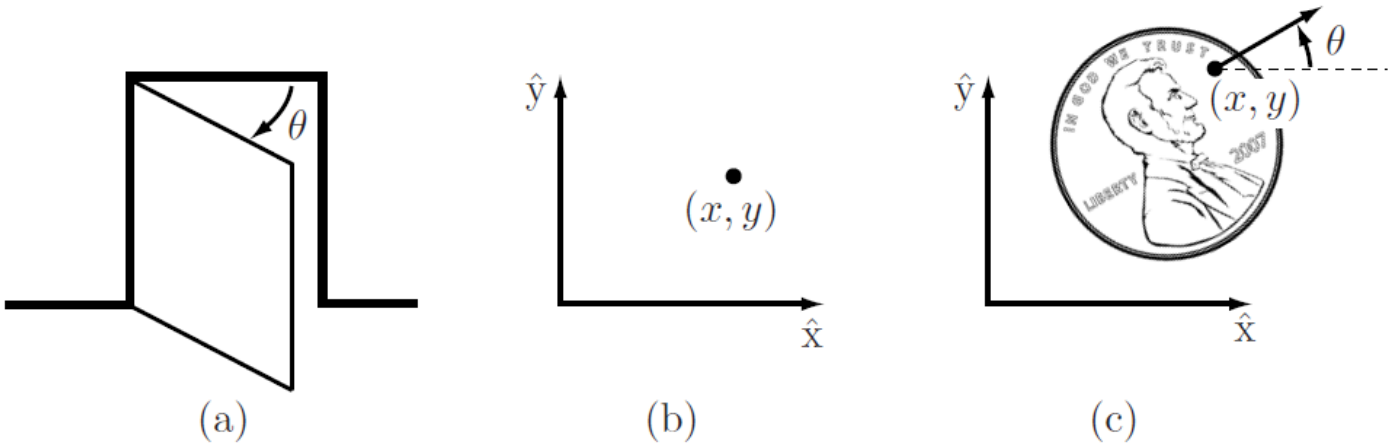


(b)



(c)

Degrees of Freedom (dof)



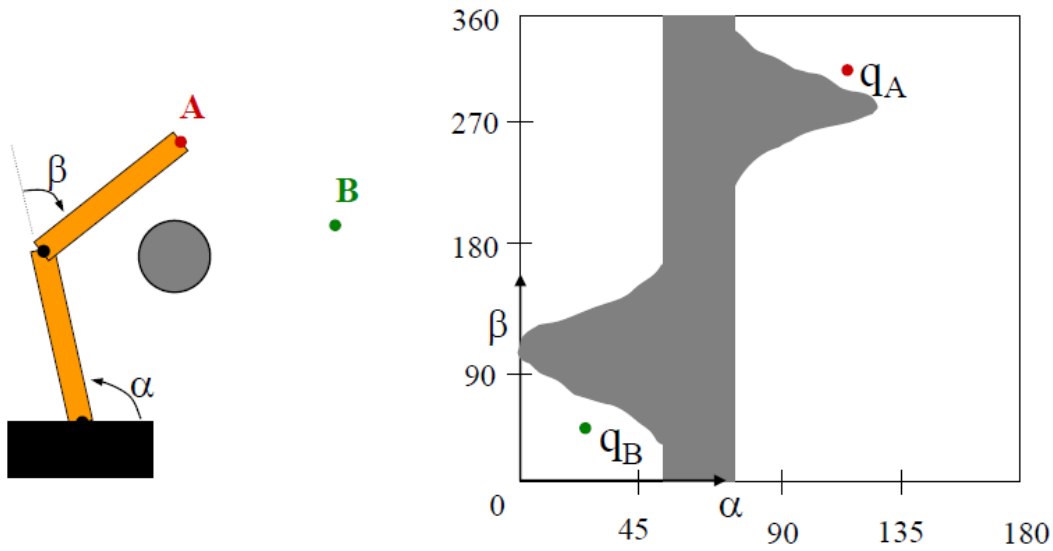
The configuration of a door is described by the angle θ (b) The configuration of a point in a plane is described by coordinates (x, y). (c) The configuration of a coin on a table is described by (x, y, θ), where θ defines the direction in which Abraham Lincoln is looking.

Configuration space, work space and task space

We are going to consider three space representations:

- The **workspace** is a specification of the robot's possible positions in the environment.
- The **configuration space (C-space)** is a specification of the robot's attainable positions in the environment. It is the n-dimensional space containing all possible configurations of the robot.
- The **task space** is the space in which the robot's task can be naturally expressed.

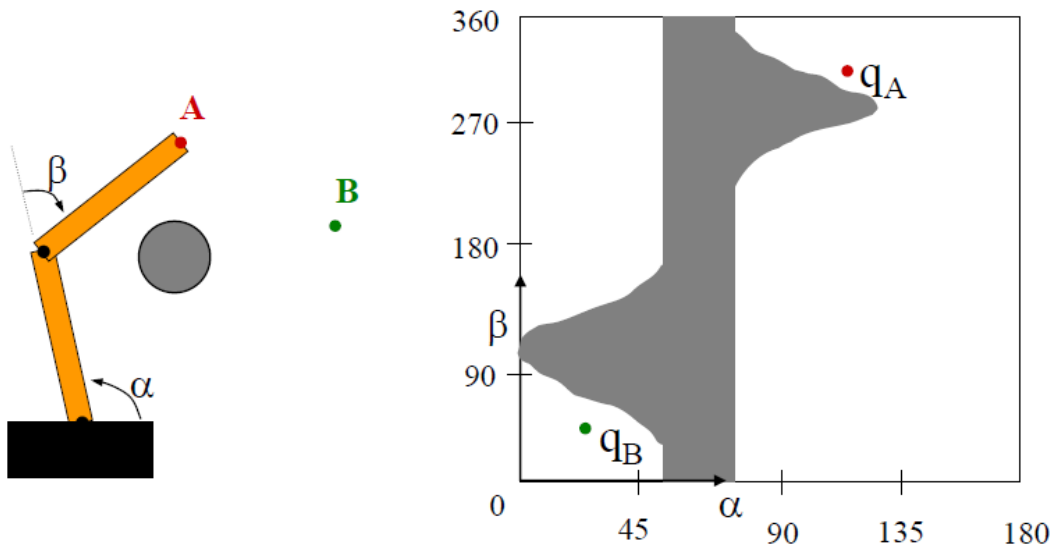
Configuration space, work space, and task space: example



- This robot arm has two joints that move independently.
- Moving these joints alters the coordinates of the elbow and the gripper.

What's the workspace, configuration space and task space here ?

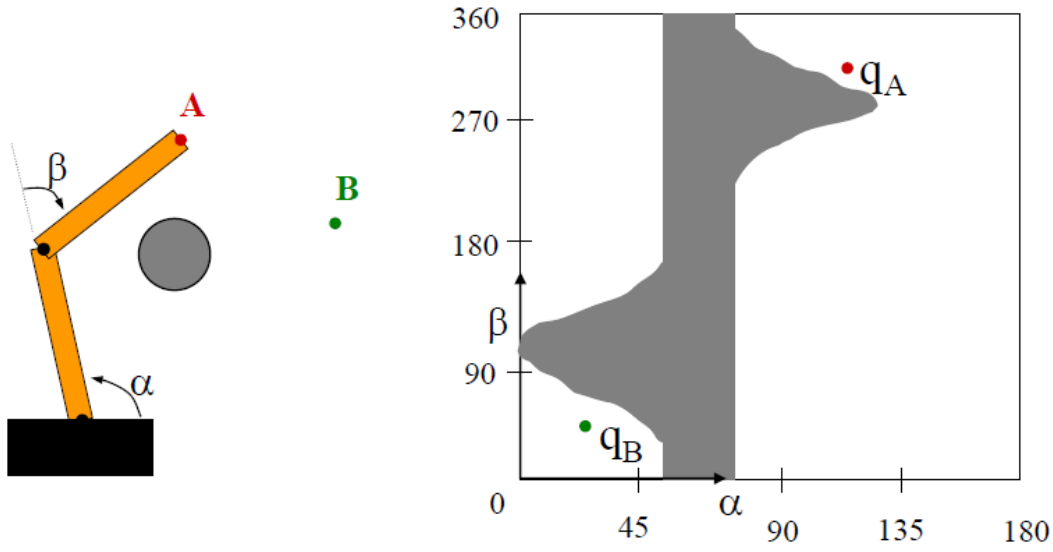
Configuration space, work space, and task space



The **workspace** is a specification of the configurations that the end-effector of the robot can reach.

Here, the **workspace** is defined by the (x,y) coordinates of the robot that are specified in the same coordinate system of the world it's manipulating (left image).

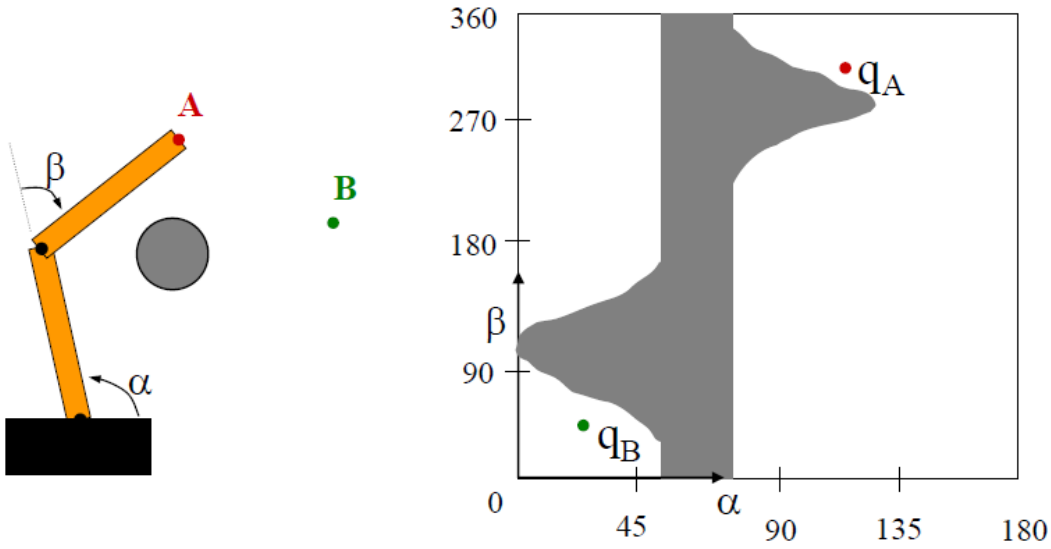
Configuration space, work space, and task space



- The configuration space (C-space) $\mathcal{C}$ is the n-dimensional space containing all possible assignments to the DOFs of the robot.
- Here, the C-space is defined by α and β and is represented in the right figure.

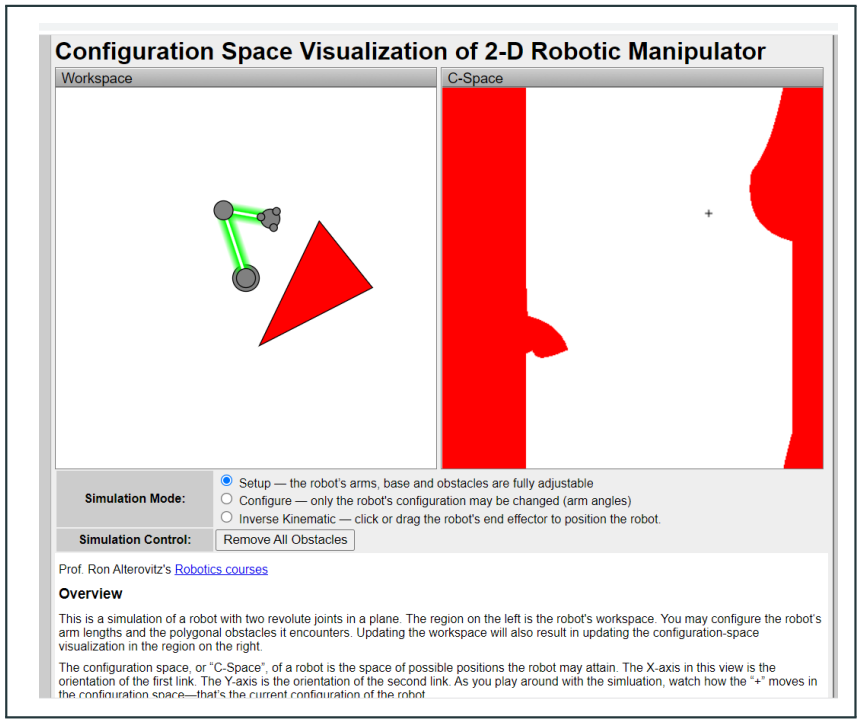
Can the robot get from A to B ?

Configuration space, work space, and task space



- The **task space** is a space in which the robot's task can be naturally expressed.
- For example, if the robot's task is to plot with a pen on a piece of paper, the task space would be R^2.
- The decision of how to define the task space is driven by the task, independently of the robot.

Configuration space, work space, and task space



https://courses.grainger.illinois.edu/cs440/fa2025/lectures/robot-simulator/simulator.html

https://www.youtube.com/watch?v=SBFwgR4K1Gk

Summary

- Basic components of a robot
- Defining the state of the robot

What next ?

- 3-layered decision making in robotics

